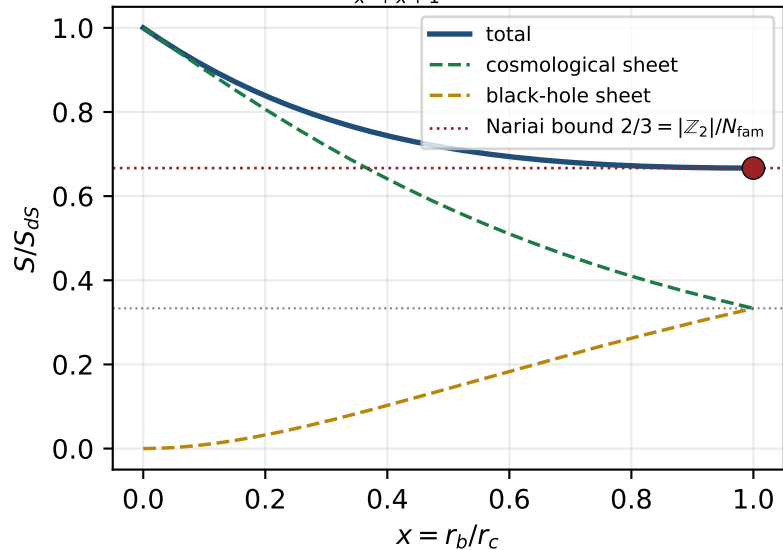


(a)  $S_{\text{tot}}/S_{\text{dS}} = \frac{x^2+1}{x^2+x+1}$  (denominator =  $\Phi_3$ )



(b) Koide-form quotient of the three roots:  
range  $[3/8, 1/2]$  = the nonzero  $SU(4)_1$  weights

